

Data Analytics (DA)

Module – 2

Power BI Assignment 2 – DAX & Data Visualization

Project Title

Sales Performance Analysis Dashboard using Power BI and DAX

Project Objective

The objective of this project is to analyze sales data using Microsoft Power BI by implementing data modeling, DAX calculations, and interactive visualizations. The dashboard helps in monitoring sales performance, comparing sales against targets, analyzing profitability, identifying top-performing product categories, and understanding geographic sales distribution.

The project focuses on:

- Data Import and Preparation
- Data Modeling and Relationships
- DAX Calculated Columns
- DAX Measures
- KPI Reporting
- Sales Trend Analysis
- Geographic Analysis
- Category-wise Performance Analysis
- Interactive Dashboard Creation

Dataset Description

The project uses multiple datasets containing sales transactions and sales targets.

- **Dataset 1: Orders Data**
Contains transactional sales information.
- **Dataset 2: Order Details**
Contains detailed order-level sales records.
- **Dataset 3: Sales Target**
Contains category-wise target values.
- **Dataset 4: Total Target by Month**
Contains monthly sales target information.

Column Description

Column Name	Description
Order ID	Unique identifier for each order
Order Date	Date when order was placed
Customer Name	Customer purchasing the product
Category	Product category
Sub-Category	Product sub-category
City	Customer city
State	Customer state
Amount	Sales amount
Profit	Profit generated
Quantity	Quantity sold
Profit Margin	Profit percentage
Target	Sales target value

Data Cleaning & Transformation

The following transformations were performed:

Data Validation

- Verified column names and data types.
- Removed inconsistencies from imported datasets.

Data Modeling

Relationships were established between:

- Orders Data
- Order Details
- Sales Target
- Total Target by Month

DAX Calculated Columns

1. Revenue per Order

Revenue per Order =

'Order Details'[Amount] * 'Order Details'[Quantity]

Purpose:

Calculate revenue generated from each order.

2. Sales Category

Classifies sales records into:

- Above Average
- Below Average

Purpose:

Identify high-performing and low-performing sales transactions.

DAX Measures Created

- **Total Amount:** Calculates total sales amount.
- **Total Profit:** Calculates total profit generated.
- **Total Quantity:** Calculates total quantity sold.
- **Order Count:** Calculates total orders placed.
- **Max Profit:** Identifies maximum profit generated.
- **Average Profit in Delhi:** Calculates average profit for Delhi city.
- **Target by Month:** Calculates monthly sales targets.

Page 1 – Core Dashboard

Visualizations

1. Sales Target Achievement by Category

Compares actual sales amount against target values for each category.

Insight

- Technology category achieved strong sales performance.
- Clothing category showed higher target values compared to actual sales.

2. Monthly Sales Trend

Displays sales trend over time.

Insight

- Sales remain consistent throughout the available period.
- Useful for monitoring monthly business performance.

3. Max Profit by Sub-Category

Displays profit contribution by each sub-category.

Insight

- Machines and Bookcases contributed higher profits.
- Certain sub-categories generated lower profit percentages.

4. Profit vs Quantity by Sub-Category

- Shows relationship between quantity sold and profit earned.

Insight

- Higher quantity products generally generated higher profits.
- Some low-quantity products produced significant profits.

5. Total Sales Amount KPI

- Value Displayed: **3.89 Million**

Insight

- Represents overall sales generated during the analysis period.

6. Total Sales Target KPI

- Value Displayed: 4.09 Million

Insight

- Shows overall business sales target.

Type of Analysis

Descriptive Analysis

- Total Sales Amount
- Total Sales Target
- Monthly Sales Trend

Diagnostic Analysis

- Sales Target Achievement by Category
- Max Profit by Sub-Category

Predictive Analysis

- Profit vs Quantity by Sub-Category helps identify future profitable categories.

Prescriptive Analysis

- Focus marketing and inventory investments on high-profit sub-categories.
- Improve sales efforts in categories with lower target achievement.

Page 2 – Advanced Analysis

Visualizations

Minimum Target by Segment

- Displays lowest target values across categories.

Insight

- Technology category has the minimum target value.
- Helps identify low-target business segments.

Sales Performance Matrix

- Displays monthly sales and targets together.

Insight

- Enables comparison of actual performance against monthly goals.
- Useful for management-level performance tracking.

Type of Analysis

- **Descriptive Analysis:** Monthly and category-wise performance overview.
- **Diagnostic Analysis:** Identifies categories missing target expectations.
- **Predictive Analysis:** Historical trends can be used for future target planning.
- **Prescriptive Analysis:** Adjust future targets and improve category-level sales strategies.

Page 3 – Geographic & Distribution Analysis

Visualizations

Geographic Sales Analysis

- Displays sales distribution by city.

Insight

- Sales activity is distributed across multiple regions.

- Helps identify major sales locations.

Sales Distribution by Sub-Category

- Treemap showing contribution by product sub-category.

Insight

- Machines and Bookcases dominate overall sales contribution.
- Helps identify top-performing products.

Order Count by State

Displays order volume by state.

Insight

- Order distribution is spread across multiple states.
- Useful for regional sales planning.

Type of Analysis

- **Descriptive Analysis:** Regional sales distribution.
- **Diagnostic Analysis:** Identify states with lower order volumes.
- **Predictive Analysis:** Forecast demand by location.
- **Prescriptive Analysis:** Increase marketing efforts in low-performing regions.

Conclusion

- This project successfully demonstrates the implementation of Power BI Data Visualization and DAX concepts. Data from multiple sources was modeled using relationships, transformed using calculated columns and measures, and presented through interactive dashboards.
- The final solution provides valuable business insights into sales performance, profitability, target achievement, product contribution, and geographic distribution. The dashboard enables data-driven decision-making and serves as an effective business intelligence reporting solution.